

Week 3 Lab 2

Part 1

1) Write a program that tosses 50,000,000 random (x, y) points into a 1×1 square to approximate π ($\pi \approx 4 \times \frac{\text{hits}}{\text{total}}$)

The more points there are, the more accurate the estimate. With 50,000,000 points, the margin of error should be very small (~ 0.0001 – 0.001), meaning the value of π should come out to approximately 3.1415–3.1417.

```
PS C:\Users\GUN\Downloads\practice_1\lab_2_week_3> javac Part1.java
PS C:\Users\GUN\Downloads\practice_1\lab_2_week_3> java Part1
totalHits = 16516479 / 50000000
pi ~= 1.321318
time = 240.5 ms
```

here we see that pi is correct.
but in task was wrote that

Part 2

Part 3